

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Even Semester)

Innovative Practices Description

Degree, Semester& Branch: II Semester B.E. ECE B

Course Code & Title: BE 8254 Basic Electrical and Instrumentation Engineering

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic: Three phase power supply, Star connection, Delta Connection –
Balanced Load

Name of the Innovative Practice: Minute Paper

Date : 08.01.2020

ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

Description of One Minute Paper Activity: A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day’s lesson and provides the instructor with useful feedback. The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the instructor's main idea and what the students perceived as the main idea are the same.

Goals (Learning Outcomes):

1. The students will be able to acquire better knowledge about Three phase power supply, Star and Delta connection.
2. The students will be able to understand the difference between star and delta Connection.

Use of appropriate methods:

Justification for choosing the One Minute Paper Activity:

Already handled three phase power supply, Star connection, Delta connection in previous class. However, I have reviewed about it. I want to know the knowledge level of the students about three phase power supply, Star connection, Delta connection. For that, I have chosen One Minute Paper as an Innovative practice method. In that I have asked some questions related with three phase power supply.

Effective presentation Details of the Implementation:

S.No.	Date	Topic(s)	Activity
1.	08.01.2020	Three phase power supply, Star connection, Delta connection – Balanced Load	Minute Paper

Phase sequence :- A-B-C
 There are three phase system
 positive sequence (RYB)
 Negative sequence
 Interconnections of windings :-
 independent connection
 star connection
 Delta connection
 Line voltage :-
 $V_L = V_R = V_Y = V_B$
 Phase voltage
 $V_{PH} = V_R, V_Y, V_B$
 active power = $P = V_{ph} I_{ph} \cos \phi$
 reactive power = $Q = V_{ph} I_{ph} \sin \phi$
 Apparent power = $S = V_{ph} I_{ph}$
 Three unbalanced impedances
 $Z_1 = Z_1 \angle \phi_1$
 $Z_2 = Z_2 \angle \phi_2$
 $Z_3 = Z_3 \angle \phi_3$

Phase sequence
 1) +ve phase sequence
 2) -ve "
 +ve $\rightarrow$ clockwise RYB with an angle difference 120°
 -ve $\rightarrow$ anticlockwise RYB.
 Star Connection
 $I_L = I_{ph}$
 $V_L = \sqrt{3} V_{ph}$
 Delta Connection
 $V_L = V_{ph}$
 $I_L = \sqrt{3} I_{ph}$
 Balanced Star Connection
 $Q = V_{ph} I_{ph} \sin \phi$
 Total power = $3 V_{ph} I_{ph} \sin \phi$
 $S = \sqrt{3} = 3 V_{ph} I_L \sin \phi$



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was assessed by evaluating the one minute paper and its feedback was given to the students in the next class.
- The assessment of effectiveness of the activity was felt when most of the students eagerly participated and they have written power factor, total power, active power, reactive power and star- Delta conversion.

Reflective Critique:**Benefits:**

The students confused about the unbalance and unbalance load, through minute paper they clarified the difference between balanced load and unbalanced load. Then students realized Star-Delta conversion and most of the students have answered properly in the one minute paper.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	1	-	1	-	1

Reference:

- S.K.Bhattacharya ,“Basic Electrical and Electronics Engineering”, Pearson India, 2011.
- [Microsoft Word - tips template minute paper.doc \(tufts.edu\)](#)